Raising the Floor or Changing the Test?
AI-Assisted Course Design and Student Outcomes in Two Community College Mathematics Courses

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Abstract

There is little evidence on what happens to students when instructors use generative AI to build course materials. I compared two terms of each of two mathematics courses I taught at a large California community college, introductory statistics and college algebra, each taught once before and once after I used AI to design the materials ($N = 106$ students who took the final exam). In algebra, where AI helped me build scaffolded, aligned materials, the average final exam score rose from 61.8% to 78.7% among students who took it, with the largest gain at the bottom: the share below 50% fell from 31% to 5%. Fewer algebra students took the final with AI, and the algebra pass rate among all enrolled students was about the same. The algebra final also changed from free response to mostly multiple choice; with the points a student could earn by guessing removed, the gains shrink and are no longer clear. In statistics, where AI helped me make the exams more demanding, scores held steady. In both AI terms, homework scores crowded toward 100%, and so did the statistics quizzes; quizzes built together with the exams kept a link to the final, while question-bank quizzes showed almost none, in a comparison between courses that already differed before AI and too small to settle. These are associations from one instructor's small classes, not proof.

Keywords: generative AI, course design, community college, mathematics, formative assessment, constructive alignment, equity

1 Introduction

Generative AI has entered college classrooms from two directions. Instructors use it to draft and revise course materials, and students use the same tools, allowed or not, to complete their work. Research has followed the same split. Studies of instructors' use describe the materials AI produces (Kasneci et al., 2023; Mollick & Mollick, 2023), and studies of students' use

focus on academic integrity (Cotton et al., 2024; Newton & Essex, 2024). The question instructors ask first is rarely studied: when a course is redesigned with AI, what happens to the students?

When outcome evidence exists, it is usually a difference in averages. An average can hide what matters most at an open-access college: which students moved. A 17-point gain that comes from lifting the weakest students out of failure means something different from the same gain coming from strong students scoring higher. Looking at the whole distribution of scores, not only the average, has a long history in economics (Koenker & Bassett, 1978) but is nearly absent from research on AI in teaching.

This paper is a practitioner case study. I taught four sections of two courses at one community college: introductory statistics and college algebra, each once with conventionally designed materials and once with materials I designed with AI. The two courses let me ask whether "the effect of AI" is one thing at all. In statistics I used AI to make the exams more demanding; in algebra I used AI to scaffold the work and to build every piece of the course to fit together. A companion paper analyzes the exams themselves and reports the same classes' average final exam scores and pass rates (Dabagh, 2026). Here I ask what happened to the students, in three plain questions.

1. **After I redesigned each course with AI, did students score higher or lower on the proctored final exam, and did as many take it?**
2. **Which students gained: the weakest, the strongest, or everyone?**
3. **Did homework and quiz scores still tell students, and me, who was ready for the final?**

I was the instructor in every section and assigned every grade, so this design cannot separate the AI redesign from my own growth as a teacher or from differences between classes. I describe what happened, not what caused it.

2 Background

2.1 AI as an instructor's design tool

Reviews of AI in higher education find mostly student-facing tools, such as tutoring and writing support, and few studies of instructors using AI to design courses (Lo, 2023; Zawacki-Richter et al., 2019). Practitioner accounts argue that large language models can draft items, examples and scaffolds that an instructor then reviews (Denny et al., 2024; Kasneci et al.,

2023; Mollick & Mollick, 2023). What is missing is evidence linking that use to what students then do on exams that AI did not grade and students could not hand off: proctored exams.

2.2 The hope: more time for every student

The hope behind AI in the classroom is about time. If AI can take over part of the routine work of preparing materials and feedback, and give students more explanations and practice than one instructor can write, instructors may have more time to help students (Kasneci et al., 2023; U.S. Department of Education, 2023). At an open-access college, where preparation in one room varies widely, the hope is that this time reaches the students who need it most. It is not a guarantee: the same tools may deepen existing gaps in access to technology and learning (UNESCO, 2023), carry the biases of their training data, and invite students to lean on them in place of learning (Kasneci et al., 2023). As one voice in the federal report put it, "AI brings educational technology to an inflection point. We can either increase disparities or shrink them, depending on what we do now" (Shilling, as cited in U.S. Department of Education, 2023, p. 7). This study cannot test the hope directly, but it can show what happened to the students at the bottom of the distribution when one instructor used AI to rebuild two courses.

2.3 Alignment and formative assessment

Two older ideas predict where any effect should appear. Constructive alignment holds that students learn best when goals, activities and assessments fit together (Biggs, 1996; Biggs & Tang, 2011). Formative assessment helps only when it gives honest information about readiness, to the student and to the teacher (Black & Wiliam, 1998). Practice that asks students to retrieve what they learned improves later retention (Roediger & Karpicke, 2006), and scaffolded materials help beginners most (Sweller, 1988). Together these ideas give a testable prediction: AI-assisted design should help through coherence, and quizzes left outside that design should lose, not just fail to gain, their value as a signal. Building a fully aligned course has always been good practice and too much work for one instructor; one-to-one support has long been known to be powerful and too costly to give every student (Bloom, 1984; VanLehn, 2011). AI changes how much work that takes.

2.4 The open-access setting

Community colleges in the United States admit every applicant and serve students of every age, background and level of preparation (Bailey et al., 2015). California's placement reforms,

Assembly Bills 705 and 1705, moved most students directly into transfer-level mathematics with support taken alongside the course, which widened the range of preparation in a single classroom (Cuellar Mejia et al., 2020, on AB 705). A randomized trial in New York had found that students assessed as needing remediation could pass a college-level course with support (Logue et al., 2016). In these classrooms the bottom of the score distribution is the policy question. Guidance for these courses stresses reasoning over recall in statistics (Chance, 2002; GAISE College Report ASA Revision Committee, 2016; Garfield & Ben-Zvi, 2008; Wild & Pfannkuch, 1999) and scaffolded, universally designed materials in every course (CAST, 2018). Whether AI-assisted design helps the students these reforms brought into the room has not been tested.

2.5 Unproctored work in the AI era

Unproctored online work can now be completed with AI, a systematic review finds widespread cheating on unproctored online exams (Newton & Essex, 2024), and generative AI makes outside help easier (Cotton et al., 2024). I treat the result mainly as a measurement problem rather than a moral one: when homework and quiz scores crowd toward 100%, they stop telling anyone who is ready.

3 Method

3.1 Setting and design

The study compares intact classes, so it is a quasi-experiment with no random assignment. The college is a large urban community college in Southern California, with about 23,000 students in Fall 2024, about 40% of them Latinx (National Center for Education Statistics, n.d.). I taught all four sections, each with a corequisite support section: introductory statistics (STAT C1000, a six-week winter session; Winter 2025 before AI, Winter 2026 with AI) and college algebra for students heading to science, technology, engineering and mathematics (MATH 4, a sixteen-week semester; Spring 2025 before AI, Spring 2026 with AI).

3.2 Students and data

The data are the gradebook records of the students who took the final exam: 28 and 27 students in statistics, and 32 and 19 in algebra (106 in all). Section enrollments (29, 27, 34 and 26) are the denominators for exam taking and pass rates. The records come from the learning management system's own category scores, which apply my grading rules (dropped

lowest scores, bonus points, extra credit) as students experienced them. Each student has a study code; the analysis file contains no names or student ID numbers. The records include no background measures beyond algebra attendance: no repeat status, placement or prior courses, share enrolled in the support section, class schedule or demographics. Adjusting for attendance in algebra is the only check on differences between classes that the data allow.

3.3 What changed with AI

In the AI terms I used a large language model (Claude, Anthropic) in a structured workflow to build the course materials every student used, and I reviewed and approved every item. In the earlier terms I prepared the materials, including the final exams, in my usual way without AI assistance (Dabagh, 2026). The two AI terms differed in one important way.

- **Statistics** (my first AI term). I used AI for the exams, lectures, formula sheets and study guides, with the goal of more demanding exam questions: the share of higher-order multiple-choice questions on the final rose from 18% to 48% (Dabagh, 2026). Both the unit exams and the final were built this way. Quizzes still came from the question banks of MyOpenMath, a free online homework platform, so they were not built together with the exams.

- **Algebra** (my second AI term). I also used AI for the quizzes, built to match the unit and final exams, along with step prompts printed under each exam problem, answer choices based on common mistakes, a formula sheet, two parallel exam forms, and a preparation package with a study guide and a practice final.

In both AI terms I also revised examples, scaffolds and review materials during the term in response to what the students in front of me needed, something that takes too long for one instructor working alone. Students in the AI terms could also use an open textbook and website I wrote; use was optional and not tracked, so I do not count it as part of the change. I did not sanction student use of AI, and I did not measure it. Table 1 describes the four final exams.

Table 1: The four final exams.

	Statistics		Algebra	
	Before AI	With AI	Before AI	With AI
Questions	41 (34 multiple choice, 7 fill-in)	25 multiple choice	20 free response	20 (19 multiple choice, 1 proof)
Total points	185	100	100	100
Allowed reference	Own notes	Formula sheet, z and t tables, up to 5 pages of own notes	Own notes	Formula sheet only
Parallel forms	No	No	No	Yes (two)
Step prompts printed on the exam	No	No	No	Yes, per problem
Bonus points	None	None	None	None

Multiple-choice items had four answer choices on both finals with AI and two to four on the earlier statistics final. The cover of the earlier algebra final says "20 multiple choice questions," but every item is free response; the cover of the earlier statistics final lists 33 multiple-choice items, but the exam has 34. On the algebra final with AI, each problem printed step prompts for students to complete, students had to show their work, and credit was partial for a correct method; a correct letter with no work could receive reduced credit. The one proof was free response.

3.4 What I measured

Main outcome. The proctored final exam: unlike the unit exams, which were also proctored, it has no bonus points, and every student in the data took it. Everything else is secondary: homework, quizzes, unit exams and the overall course grade, all as percentages.

Other measures. Algebra attendance is the gradebook attendance score, based on sessions attended and including the gradebook's adjustments, so it runs from below 0 to above 100. The statistics gradebook kept attendance only as a graded category with bonuses, so it is not used. A score is *at the ceiling* when it is 95% or higher. The *practice gap* is the average of a student's homework and quiz scores minus the final exam score. *Finishing* means taking the final exam; *passing* means a course grade of 70% or higher, and I also report 60% because colleges differ on which counts.

3.5 How I analyzed the data

For each course I compared the before and with-AI classes on all ten measures (five per course) with Welch's t test, which allows different spreads in the two classes, and the Mann-Whitney test, which uses ranks, and I report Hedges' g as a standardized size. Because ten

comparisons raise the chance of a false alarm, I used the Holm correction across all ten (Holm, 1979). To see where in the distribution scores moved, I compared percentiles (the 10th, 25th, median, 75th and 90th) with bootstrap 95% confidence intervals (CI) from 10,000 resamples. Because fewer algebra students took the final with AI, I checked whether that alone could explain the gain with a trimming bound (Lee, 2009): I compared the AI class with only the top-scoring students of the earlier class, as many as would match the lower exam-taking rate. For the link between quizzes and the final I used rank (Spearman) correlations, which depend less on how scores bunch near 100%. For algebra I also adjusted for attendance. Pass and finishing rates use every enrolled student and Fisher's exact test. Because the algebra final changed from free response to mostly multiple choice, I also removed the points a student could earn by guessing: 19 four-choice questions worth 5 points each earn 23.8 points on average from blind guessing, so I rescaled the AI scores to put that score at zero and 100 at 100. The percentile intervals, the trimming bound, the correlations and this guessing check are secondary analyses and are not corrected for multiple comparisons. Details are in the supplement.

Software and AI use. All analyses were run in Python 3.13.2 with pandas 3.0.3, NumPy 2.2.5, SciPy 1.17.1, statsmodels 0.14.6 and Matplotlib 3.10.3. One script produces every number, table and figure from the data. Generative AI (Claude, Anthropic) was used to assist with writing, running, and checking the analysis code, checking references, and drafting, editing, and proofreading the text. I designed the study, directed and reviewed all of this work, and take full responsibility for the content.

3.6 Ethics

I asked the college for institutional review board review of my research on student outcomes in my own sections. At the time, the college was not conducting full institutional review board reviews because of budget and staffing limits. Its Office of Institutional Research advised that the work could proceed as educational research with the approval of the department chair, and the chair approved it (March 2026). No approval or exemption was obtained from any review board; departmental approval is not ethics review, and I do not present it as such. The study analyzes records of ordinary teaching after the fact, with no research intervention and no contact with students for research purposes; no consent was sought, because no student was contacted. Results are reported for groups; the figures show individual scores with no identifying information. In considering the ethics of the study, I weighed three issues: privacy, handled by an analysis file with no names or ID numbers and by reporting results for groups; the absence of any research intervention or contact with students; and my dual

role as instructor and researcher, which I disclose here and treat as a limitation.

4 Results

4.1 Finding 1: algebra final exam scores rose; statistics scores held on a harder exam

Table 2 gives all ten comparisons and Figure 1 shows every student's final exam score. Two results remain clear after correcting for ten comparisons: the algebra final exam and statistics homework.

Algebra. The average final exam score rose from 61.8% to 78.7%, about 17 points (95% CI 5.5 to 28.3; Holm $p = .040$; $g = 0.75$). Scores were also less spread out (standard deviation 25.3 before, 15.2 with AI). Average homework and attendance scores did not clearly differ. Adjusting for attendance leaves the gain at 17.4 points ($p = .009$), and the gain appears in every third of attendance: 24.8 points among the lowest attenders, 16.7 in the middle and 16.3 among the highest. The algebra final also changed from free response to mostly multiple choice; with guessing points removed the average gain is 10.3 points (95% CI -2.6 to 23.2) and no longer clear (Finding 2).

Statistics. The average final exam score was 74.0% before and 77.6% with AI, not a clear change ($p = .456$), even though the AI exam was much more demanding (Dabagh, 2026). Homework rose by 14.4 points. Unit exam scores fell by 9.9 points and spread out much more; the unit exams were rebuilt at the same higher demand as the final, so this fits harder exams separating students more sharply. The overall grade and pass rates did not clearly change.

Table 2: Every comparison among students who took the final exam (with AI minus before).

Course	Measure	Before	With AI	Change	95% CI	p	Holm p	Mann-Whitney p	g
Statistics	Homework*	82.1	96.5	+14.4	[+6.2, +22.6]	.001	.012	.001	+0.93
	Quizzes	85.2	92.4	+7.1	[+0.3, +14.0]	.041	.245	.005	+0.56
	Unit exams	89.8	80.0	−9.9	[−18.1, −1.6]	.020	.142	.097	−0.66
	Final exam	74.0	77.6	+3.6	[−6.0, +13.2]	.456	> .99	.259	+0.20
	Overall grade	84.7	84.5	−0.2	[−6.6, +6.2]	.956	> .99	.755	−0.01
Algebra	Homework	74.0	79.5	+5.5	[−11.7, +22.7]	.522	> .99	.140	+0.19
	Quizzes	66.9	87.3	+20.4	[+5.0, +35.9]	.011	.085	.064	+0.66
	Unit exams	79.8	85.4	+5.7	[−5.0, +16.4]	.288	> .99	.121	+0.33
	Final exam*	61.8	78.7	+16.9	[+5.5, +28.3]	.004	.040	.025	+0.75
	Overall grade	74.2	82.6	+8.4	[−0.8, +17.7]	.073	.366	.096	+0.52

Scores are percentages; change is in percentage points with Welch 95% confidence intervals. p: Welch's t test. Holm p: corrected for all ten comparisons; * remains below .05. Mann-Whitney p: rank test. g: Hedges' g. Statistics: 28 and 27 students (one student with AI has no homework score); algebra: 32 and 19.

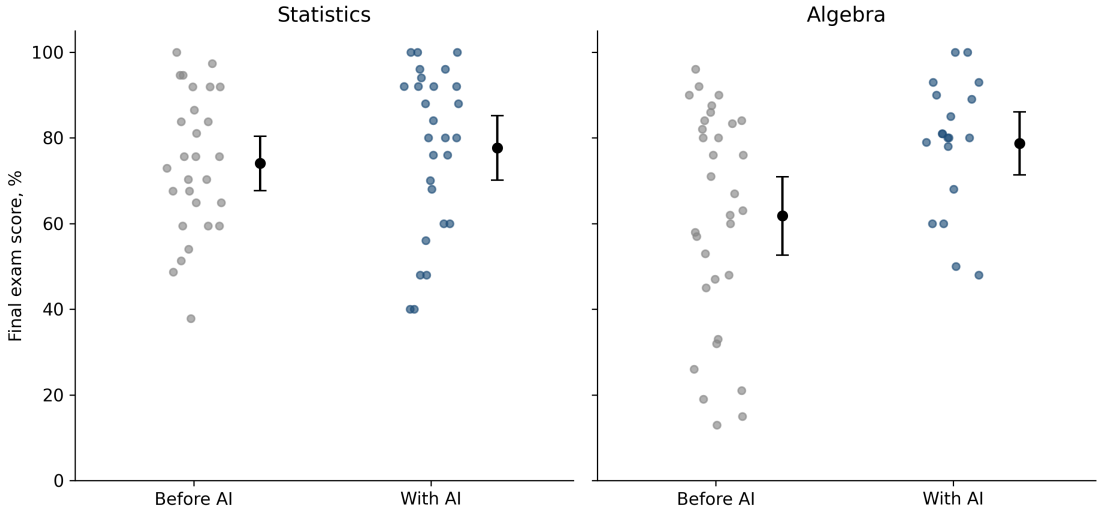


Figure 1: Final exam scores in all four classes. Dots are students; black points and bars are the average with its 95% confidence interval.

4.2 Finding 2: the algebra gain was largest at the bottom, but the exam format also changed

The algebra gain was concentrated among the students who were struggling most (Table 3, Figure 2). The 10th percentile rose from about 22 to 58, and the 25th from about 47 to 73, while the 90th barely moved (90 to 94). In plain numbers, 10 of 32 students scored below 50% before, and 1 of 19 did with AI. In statistics the bottom spread a little and the top rose a little (10th percentile 53 to 48; 90th 95 to 98), which fits a harder exam.

Fewer algebra students took the final with AI (Finding 5), so could the gain simply reflect which students sat the exam? Three things argue against it being the whole story. Average homework and attendance did not clearly differ. The gain held after adjusting for attendance and in the lowest attendance group. And the trimming bound still favors the AI class. It assumes the worst case, that the students who skipped the final were exactly the weakest, and compares the AI class with the top 25 of the 32 students of the earlier class (the upper trim of Lee, 2009). The AI class still had a higher average (78.7% against 72.7%), a higher 10th percentile (58 against 50) and fewer scores below 50% (5% against 12%). Under this worst case the average gain shrinks from 16.9 to 6.0 points (95% CI −3.5 to 15.4): the direction holds, but the size is no longer clear.

The exam itself also changed, and this is the strongest rival explanation. The earlier algebra final was 20 free-response problems; the AI final was 19 four-choice problems and one proof (Table 1). On free response, a student who cannot start a problem earns nothing for it; on multiple choice, guessing alone earns some points, which lifts the lowest scores most. Removing the average score from blind guessing, 23.8 points, shrinks the gains (Table 3): the 10th percentile rises by 23 points instead of 36, the average by 10.3 (95% CI −2.6 to 23.2), and the share below 50% falls to 21% rather than 5%. None of these changes is clear, although the pattern, largest at the bottom and small at the top, remains. This likely removes more than blind guessing gave: students had to show their work, a correct letter without work could lose credit, wrong choices were built from common mistakes, and no student with AI scored at or below 23.8 points. It does not remove other help the format can give, such as ruling out choices or substituting them into an equation. The two columns of Table 3 are therefore a guide, not exact bounds, and this study cannot tell how much of the rise at the bottom came from the redesign and how much from the change of format.

Table 3: Algebra final exam scores across the distribution, as scored and with guessing points removed.

		With AI, as scored			With AI, guessing points removed		
Algebra final exam	Before	Score	Change	95% CI	Score	Change	95% CI
10th percentile	22	58	+36	[+11, +57]	45	+23	[−3, +49]
25th percentile	47	73	+26	[+3, +50]	65	+18	[−10, +44]
Median	65	80	+15	[0, +30]	74	+9	[−7, +25]
75th percentile	83	90	+7	[−4, +17]	86	+3	[−10, +15]
90th percentile	90	94	+4	[−3, +16]	93	+3	[−6, +16]
Share below 50%	31%	5%	−26	[−45, −6]	21%	−10	[−34, +14]
Average	61.8	78.7	+16.9	[+5.5, +28.3]	72.0	+10.3	[−2.6, +23.2]

Percentiles of the final exam score, in percent; 95% confidence intervals from 10,000 bootstrap resamples. Guessing points removed: AI scores rescaled as (score minus 23.8) divided by (100 minus 23.8), times 100, the average score from blind guessing on 19 four-choice problems worth 5 points; the earlier free-response exam has no guessing floor. Statistics percentiles are in the supplement.

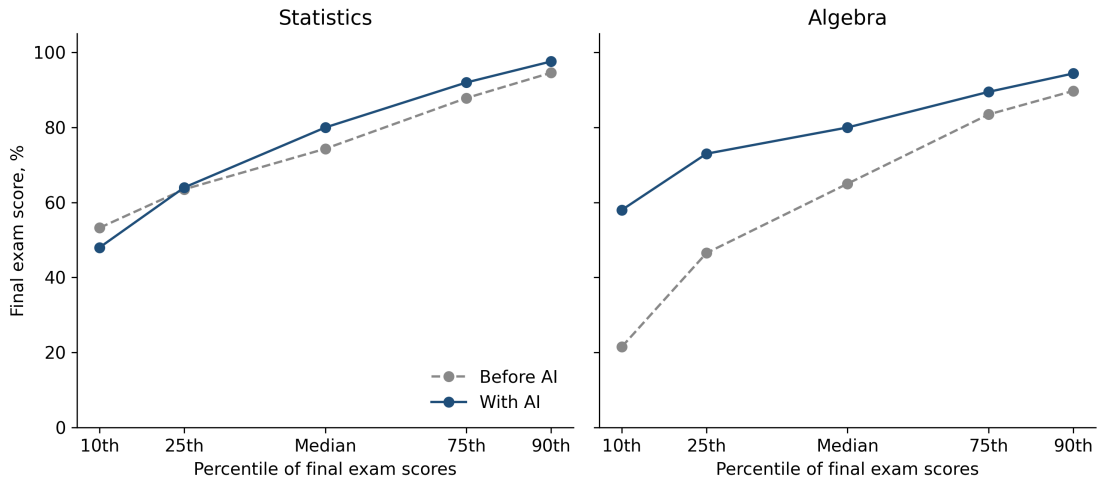


Figure 2: Final exam scores at five percentiles, before and with AI, as scored. In algebra the bottom rose most and the top barely moved; Table 3 shows how much of that remains once guessing points are removed.

4.3 Finding 3: homework and quiz scores crowded toward 100% in both AI terms

In both courses more students scored at the ceiling (Table 4). Homework at 95% or higher rose from 32% to 85% of students in statistics and from 22% to 63% in algebra; quizzes rose from 25% to 52% in statistics and barely changed in algebra (38% to 42%), though the algebra quiz average rose 20.4 points. Two explanations fit, and the data cannot separate

them: AI-designed practice may be easier to complete by design, with steps and retries, and students may have used AI on unproctored work.

4.4 Finding 4: quizzes built with the exams kept a link to the final; question-bank quizzes showed almost none

What the crowding did to the quizzes depended on how they were built. In statistics, where quizzes still came from the question bank, the rank correlation between quiz average and final exam fell from .25 to .05 (95% CI −.36 to .46): with AI, quiz scores told almost nothing about the final, and the gap between practice scores and the final widened from 9.6 to 17.0 points. The proctored unit exams still tracked the final closely (.81). In algebra, where quizzes were built together with the exams, the same correlation rose from .46 to .58 (95% CI .12 to .86), and the practice gap narrowed from 8.7 to 4.7 points (Figure 3). The two courses already differed before AI: the quiz and final correlation was .25 in statistics and .46 in algebra, and the intervals for all four correlations overlap, so the contrast is between courses as much as between quiz designs; in statistics the question-bank quizzes never had a strong link to lose. The difference between the two AI terms is suggestive rather than settled (Fisher's z test for two correlations, $p = .068$), and the algebra term was also my second AI term, so alignment and my experience cannot be separated. One more pattern points the other way: in algebra the correlation between the proctored unit exams and the final fell from .70 to .60, while in statistics it rose from .75 to .81.

Table 4: How well homework and quizzes signaled readiness for the final.

	Statistics		Algebra	
	Before	With AI	Before	With AI
Quizzes built with the exams?	No	No	No	Yes
Homework at 95% or higher, % of students	32	85	22	63
Quizzes at 95% or higher, % of students	25	52	38	42
Quiz and final exam, rank correlation	.25	.05	.46	.58
Quiz and unit exams, rank correlation	.39	.17	.46	.75
Unit exams and final exam, rank correlation	.75	.81	.70	.60
Homework and quizzes minus final, points	+9.6	+17.0	+8.7	+4.7

Students who took the final exam. Rank correlations are Spearman's ρ. The practice gap uses the students with both a homework and a quiz score.

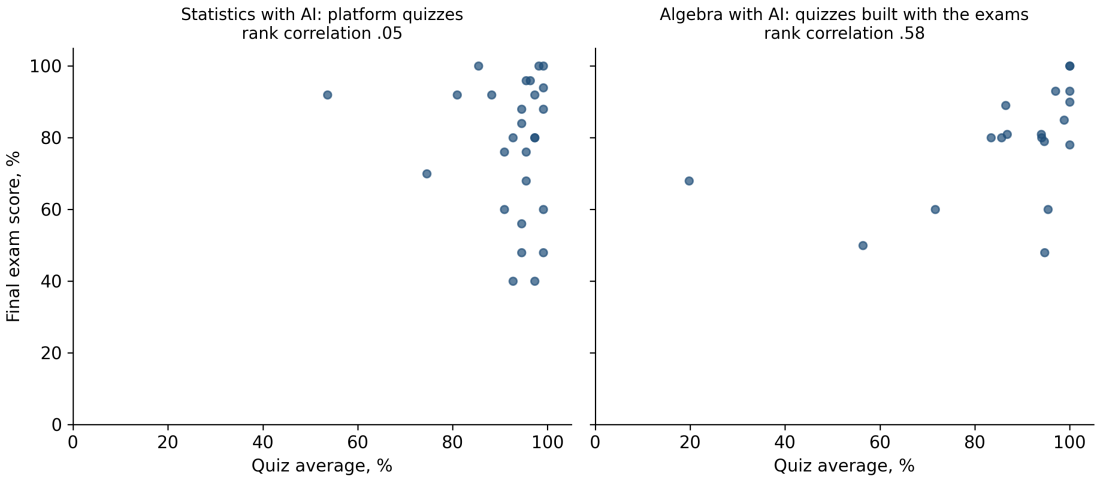


Figure 3: Quiz average and final exam score in the two AI terms. Left: statistics, quizzes from a question bank. Right: algebra, quizzes built together with the exams.

4.5 Finding 5: fewer algebra students took the final, and the pass rate held

In statistics nearly every student took the final in both terms. In algebra, 94.1% of enrolled students took the final before and 73.1% with AI (Fisher's exact test, $p = .032$; Table 5). The pass rate among all enrolled students was about the same: 52.9% and 57.7% at 70%, and 67.6% and 69.2% at 60%. Higher scores among those who took the final and fewer students taking it roughly offset each other. A single model across both courses gives 9.8 points (95% CI 1.9 to 17.7), but it averages two different changes, 3.6 points in statistics and 16.9 in algebra; the difference between them (13.3 points, $p = .095$) is not clear with classes this small.

Table 5: Taking the final exam and passing, among all enrolled students.

	Statistics		Algebra	
	Before	With AI	Before	With AI
Enrolled	29	27	34	26
Took the final exam	28	27	32	19
% of enrolled	96.6	100.0	94.1	73.1
Course grade 60% or higher, % of enrolled	93.1	92.6	67.6	69.2
Course grade 70% or higher, % of enrolled	93.1	85.2	52.9	57.7

Passing is a course grade at or above the threshold shown; students who did not take the final count as not passing.

5 Discussion

This study has three main findings. First, in algebra, where I used AI to scaffold the work and align every piece of the course, final exam scores rose by about 17 points, and the gain was largest at the bottom, though the change from free response to multiple choice may account for part or much of it. Second, in statistics, where I used AI to make the exams harder, students held steady on a more demanding exam. Third, in both AI terms homework and quiz scores crowded toward 100%, and only the quizzes built together with the exams kept telling students who was ready, in a comparison too small to settle and between courses that already differed before AI.

5.1 What this means for practice

AI-assisted design is not one intervention. The two redesigns had different goals and produced different results. A single "effect of AI" would hide both.

Look at the bottom of the distribution, and hold the exam format fixed. An average gain of 16.9 points sounds like everyone improved. Percentiles show where the change was largest: among the students most at risk of failing, the students placement reform brought into transfer-level mathematics (Cuellar Mejia et al., 2020). A change of exam format can produce the same pattern, so a redesign that also changes the format needs a check like the one in Finding 2.

Build quizzes together with the exams. When a course is redesigned with AI, practice left outside the design can stop measuring readiness. Bring quizzes into the same design as the exams; where a question bank must stay, add short proctored checks; and watch two numbers every term, the share of scores at the ceiling and the correlation between quizzes and exams. Both come from any gradebook.

Count who finishes, not only who scores. A redesign should be judged on how many students take the final as well as on their scores; here the two roughly offset in the pass rate (18 of 34 enrolled students passed before and 15 of 26 with AI).

Associations, not proof. One instructor, two courses and small classes cannot show that AI caused these results. A larger study with common final exams, recorded repeat status, measured student AI use and several instructors would.

5.2 Why the gain was largest at the bottom

The change of exam format is the first explanation to weigh, and Finding 2 shows it could account for part or much of the pattern. Two features of the algebra redesign also point

toward the bottom. Revising examples and scaffolds during the term for the students actually enrolled is, by its nature, aimed at the students who are struggling, much as one-to-one tutoring is (Bloom, 1984; VanLehn, 2011), and scaffolded materials help beginners most (Sweller, 1988). The data cannot separate these from each other or from the change of format. Both turn course materials into living documents, revised for the students in the room rather than fixed before the term begins; the AI workflow is what made that possible for one instructor.

The drop in algebra exam taking deserves care. It may reflect ordinary differences between classes of this size, students' standing before the final, or differences in chances to defer an exam; the design cannot tell. Under the worst reading, the trimming bound keeps the direction of the gain and the higher floor, but shrinks the average gain to about 6.0 points (95% CI −3.5 to 15.4), which these class sizes cannot separate from zero.

5.3 Why the quizzes split

Formative work helps only when it carries honest information about readiness (Black & Wiliam, 1998). Crowding toward 100% can come from easier-to-complete practice, from student AI use (Cotton et al., 2024; Newton & Essex, 2024), or from practice that no longer matches the exams (Biggs, 1996). The two AI terms cannot separate the first two, but they bear on the third: homework crowded in both, yet only the aligned quizzes kept their link to the final, as alignment would predict (Biggs, 1996). The algebra quizzes also crowded less at the ceiling (38% to 42%, against 25% to 52% in statistics), and the comparison is between courses that already differed and too small to settle.

5.4 What this says about the hope

The results fit the hope in direction but cannot confirm it. In algebra, I used the AI workflow to revise materials during the term for the students in front of me, and the gain, as scored, was largest at the bottom. But the exam format also changed, the classes were small, and I was the only instructor, so these data cannot show that AI-assisted design promotes equity. They show that the hope is worth testing directly: with the exam held fixed, the instructor's time measured, and results reported for the students most at risk.

5.5 Strengths and limitations

Strengths include gradebook records for every student who took the final in four classes, the same instructor in all of them, a primary outcome that is proctored and has no bonus points,

all ten comparisons reported with correction for multiple testing, and results reported across the whole distribution, with one script producing every number.

The study has seven main limitations.

1. **One instructor, intact classes.** The classes were not randomly assigned, and I taught and graded every section, so the results may reflect my own growth or differences between classes. The records hold no background measures beyond algebra attendance, and repeat-taker status could not be linked.
2. **Different exams, including a change of format.** The final exams differ within each course. The statistics AI exam was harder (Dabagh, 2026). The algebra AI exam was more scaffolded and changed from 20 free-response problems to 19 multiple-choice problems and one proof, so part of the rise at the bottom may be mechanical rather than learned; with the guessing points removed, the 10th percentile rose by 23 points and the change is no longer clear. Item-level scores, including the one proof, were not in the records, so the proof cannot be compared on its own. The exam covers also contain errors: the earlier algebra cover says "20 multiple choice questions" though every item is free response, and the earlier statistics cover lists 33 multiple-choice items though the exam has 34.
3. **Students who did not take the final.** The records do not show when the students who skipped the algebra final stopped attending, or whether they withdrew; the trimming bound in Finding 2 assumes the worst case for them.
4. **Small classes.** With 19 to 32 students per class, intervals are wide, the test of whether the two courses differ has little power, and percentiles near the tails rest on few students.
5. **Student AI use was not measured.** The crowding toward 100% cannot be split between material design and student AI use, and my mid-term revisions were documented by me, not logged independently.
6. **The courses differ.** A six-week statistics session and a sixteen-week algebra semester differ in format, content and students, and the algebra AI term was my second, so alignment is tied up with experience.
7. **Optional resources.** My open textbook and website were available in the AI terms with untracked use.

6 Conclusion

Across four classes of two community college mathematics courses that I taught, AI-assisted course design was followed by a large rise on the proctored algebra final, largest among the students most likely to fail. Fewer of the enrolled students took that final, and it changed from free response to mostly multiple choice, which could account for part or much of the rise at the bottom. In statistics, scores held steady on a harder final. In both AI terms, homework and quiz scores crowded toward 100%. Only quizzes built together with the exams kept their value as a signal, in a comparison between courses that already differed and too small to settle. These are associations from one instructor's classes, not proof that AI caused them.

The takeaway is simple. AI-assisted design was followed by fewer very low scores among the students who took the algebra final, but this study cannot say how much of that came from the design and how much from the new exam format. The next test should hold the exam format fixed, build the whole course, quizzes included, to fit together, and read results across the distribution, not only on average.

Statements

Use of AI. The course redesign studied here used Claude (Anthropic) as described in Section 3.3. For this paper, see Method, Software and AI use.

Data availability. The analysis file (study codes only, no names or ID numbers) and the analysis script are available from the author on request and will be deposited in an open repository on publication. Some cells are small, so a public file may need to be coarsened.

Ethics. See Method, Ethics. No review board approval or exemption was obtained.

Competing interests. I was the instructor of record for all four sections and assigned the grades analyzed, and I designed the course materials studied. No financial or other competing interests.

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